

A Meta-Analysis of Global SARS-CoV-2 Variant Incubation Periods: Epidemiological Insights for Continued Emergency Preparedness and Response Along the Continuum of Pandemic Phases

M. Anthony Cappello, PhD, REHS

Alexander S. Riordan, MPH

Elizabeth A. Best, MS

Megan R. Henry, MPH

Katrina K. Stadler, MS

Angela Shen, MPH

Stephanie Tack, MPH

Hannah C. Mazzotta, MPH

Insight Exposure & Risk Sciences Group

Abstract Understanding the evolution of global SARS-CoV-2 variant incubation periods provides valuable epidemiological insights for continued emergency preparedness and response along the continuum of pandemic phases. We conducted an updated meta-analysis of global COVID-19 mean and median incubation periods among several dominant SARS-CoV-2 variants. The overall pooled mean incubation period was 4.15 days (95% confidence interval (CI) [3.91, 4.39]) and the overall pooled median incubation period was 3.64 (95% CI [3.16, 4.13]). Our updated results show that SARS-CoV-2 continues to evolve through a gradual shortening of the incubation period over time. This information is important for informing current and future COVID-19 evidence-based decision-making, and is valuable for updating and streamlining public health guidance and recommendations, as well as advising future emergency preparedness and response planning.

Keywords: meta-analysis, COVID-19, incubation period, pandemic, epidemiology, emergency preparedness and response, infectious disease, continuum of pandemic phases

Introduction

Background

From the onset of the COVID-19 pandemic, SARS-CoV-2 has evolved to include a broad genetic diversity with multiple lineages producing numerous dominant variants, each with a unique and shifting transmissibility, virulence, and incubation period (Centers for Disease Control and Prevention [CDC], 2024a, 2025a; World Health Organization [WHO], 2024). Globally, while various peer-reviewed studies have assessed the average incubation period of COVID-19 since the onset of the pandemic, estimates vary based

on time period, number of participants, and geographic location (Wu, Kang, et al., 2022). Therefore, obtaining updated and reliable estimates of the incubation period for novel infectious diseases, such as COVID-19, is essential for informing infectious disease emergency preparedness and response and evidence-based public health decision-making. Continuous analysis of a novel disease's incubation period over time provides valuable insights for developing or updating case definitions, establishing quarantine and isolation periods, streamlining public health responses and disease investigations, and developing effective public health protocols

and policies (Lauer et al., 2020; Lilienfeld et al., 1994; Linton et al., 2020; McAloon et al., 2020; Nishiura, 2007; Rogers, 2020; U.S. Department of Health and Human Services, n.d.; WHO, 2020; Wu, Chen, et al., 2022; Wu, Kang, et al., 2022).

Objective

Given the rapid evolution of the COVID-19 pandemic, this meta-analysis aims to provide an updated estimate of the global mean and median incubation periods for COVID-19 (including historically dominant SARS-CoV-2 variants) using epidemiological data published after February 10, 2022, which was the most recent meta-analysis of COVID-19 incubation periods (Wu, Kang, et al., 2022) when our analysis began. Additionally, we discuss the importance of continued COVID-19 epidemiological research for effective emergency preparedness and response along the continuum of pandemic phases, particularly during the transition and interpandemic phases.

Methods

Literature Search

A literature search and meta-analysis were performed using guidance from the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) and the Meta-analysis of Observational Studies in Epidemiology (MOOSE) (Moher et al., 2009; Stroup et al., 2000). This article is written in accordance with the PRISMA reporting checklist.

We conducted a systematic literature search via the National Center for Biotechnology Information's PubMed search engine using the search parameter from February